Full Metrics — Topology-Agnostic Bottleneck-Ranking DDQN (final controller)

All metrics below are computed from the frozen final evaluation (final_learned_4of5_iter2_eval_per_cycle.csv, 3816 cycles, 8 topologies). Note: this final method is the Topology-Agnostic Bottleneck-Ranking DDQN (real Double-DQN, argmax-Q, bottleneck-aware OD ranking, selected-flow LP, nonselected ODs = ECMP). It is NOT the older Reward-Gated GNN-LPD method and uses no RandomForest and no reward gate; it is labelled accordingly rather than as 'RG-GNN-LPD'.

1. Method definition

Component	Role in the final method
GNN-LPD scorer	learned per-OD criticality score (one input signal)
Bottleneck-aware ranking	orders OD pairs by relief (demand x ECMP-flow x link-util) + 0.3 x GNN; selects top-K
Double-DQN (argmax-Q)	action policy: chooses KEEP or an optimize budget K from {50,100,200,300,500,800}
Selected-flow LP	feasible routing optimizer over the selected top-K ODs only
ECMP	fixed background routing for all nonselected ODs

2. Dataset / train / validation / test

Topology	Source	Role	Train cycles	Eval cycles	Zero-shot?
Abilene	SNDlib (real)	Seen	0-160 (of 0-2016)	2016-4032	No
GEANT	SNDlib (real)	Seen	0-160	672-1344	No
CERNET	real	Seen	0-160	200-400	No
Sprintlink	Rocketfuel (synthetic caps)	Seen	0-160	200-400	No
Tiscali	Rocketfuel (synthetic caps)	Seen	0-160	200-400	No
Ebone	Rocketfuel (synthetic caps)	Seen	0-160	200-400	No
Germany50	SNDlib (real)	Generalization	- (never trained)	0-288	Yes
VtlWavenet	TopologyZoo (synthetic caps)	Generalization	- (never trained)	0-40	Yes

Note: CERNET, Sprintlink, Tiscali, Ebone, VtlWavenet use synthetic (degree-based) link capacities in this pipeline; Abilene, GEANT, Germany50 use real capacities. OD-fraction comparisons across papers are only apples-to-apples on the same capacity model.

3. Normal-scenario metrics (per topology)

Topology	N	MeanPR	MedPR	PR>=.95	PR>=.90	MinPR	MeanDB	P95DB	MaxDB	Mean ms	P95 ms
Abilene	2016	0.9843	1.0000	87%	97%	0.8109	0.0058	0.0202	0.0510	10.7	20.4
GEANT	672	0.9983	0.9999	99%	99%	0.5848	0.0030	0.0057	0.0701	66.6	121.1
CERNET	200	0.9925	1.0000	96%	100%	0.8933	0.0002	0.0008	0.0077	46.1	120.7
Sprintlink	200	0.9960	1.0000	100%	100%	0.9600	0.0034	0.0090	0.0546	174.8	278.8
Tiscali	200	0.9522	0.9543	54%	100%	0.8928	0.0020	0.0092	0.0311	76.8	298.4
Ebone	200	0.9713	0.9806	82%	96%	0.8874	0.0003	0.0016	0.0133	3.0	33.6
Germany 50	288	0.9878	0.9989	96%	97%	0.4090	0.0098	0.0406	0.0659	212.6	285.5
VtlWaven et	40	0.9373	0.9383	0%	100%	0.9310	0.0007	0.0014	0.0017	295.9	307.3

3b. MLU and MLU/optimum

Topology	Mean MLU	P95 MLU	Max MLU	Mean MLU/opt	Max MLU/opt
Abilene	0.0506	0.0668	0.1132	1.0171	1.2331
GEANT	0.1091	0.1313	0.1940	1.0025	1.7101
CERNET	0.6338	0.8650	0.8975	1.0079	1.1194
Sprintlink	0.7134	0.8772	0.9029	1.0041	1.0417
Tiscali	0.6457	0.8690	0.8976	1.0508	1.1201
Ebone	0.7291	0.8767	0.9584	1.0304	1.1269
Germany50	12.4247	16.3146	26.1577	1.0181	2.4451
VtlWavenet	1.0391	1.0970	1.0987	1.0669	1.0741

MLU/optimum = our MLU / strict-optimum MLU = 1/PR. Values near 1.0 mean near-optimal routing.

4. Baseline comparison (same dataset/protocol)

Method	Pooled Mean PR	Status
ECMP (baseline)	0.6775	computed (see per-topology below)
Final DDQN (ours)	0.9852	computed (frozen Iter2)
OSPF / shortest-path	-	NOT rerun for the final method/protocol
Top-K by demand	-	NOT rerun for the final method/protocol
Bottleneck / sensitivity selectors	-	NOT rerun for the final method/protocol

Per-topology ECMP vs final DDQN PR:

Topology	ECMP PR	Final DDQN PR	Improvement
Abilene	0.4502	0.9843	+0.5340
GEANT	0.4016	0.9983	+0.5966
CERNET	0.8863	0.9925	+0.1062
Sprintlink	0.6906	0.9960	+0.3054
Tiscali	0.8128	0.9522	+0.1394
Ebone	0.9154	0.9713	+0.0559
Germany50	0.3441	0.9878	+0.6437
VtlWavenet	0.9187	0.9373	+0.0186

Only ECMP is recomputed on the exact final protocol; OSPF/top-K/bottleneck/sensitivity baselines were not rerun for the final controller and are therefore not claimed.

5. K sensitivity and action distribution

Ranking/K ablation (fixed K, carry-forward) — gate(blend) vs relief-only vs GNN-only:

Topology	K	Ranking	Mean PR	Mean ms
Sprintlink	500	gnn_only	0.9700	256.9
Sprintlink	500	relief_only	0.9979	259.3
Sprintlink	500	blend	0.9960	251.6
Sprintlink	800	gnn_only	0.9883	374.5
Sprintlink	800	relief_only	1.0000	380.1
Sprintlink	800	blend	0.9993	377.9
Tiscali	300	gnn_only	0.9268	206.4
Tiscali	300	relief_only	0.9236	193.5
Tiscali	300	blend	0.9243	207.8
Germany50	200	gnn_only	0.9356	127.6
Germany50	200	relief_only	0.9900	141.1
Germany50	200	blend	0.9555	131.1
CERNET	50	gnn_only	1.0000	55.1
CERNET	50	relief_only	1.0000	54.5
CERNET	50	blend	1.0000	61.0

Action distribution by topology (per-cycle argmax-Q choices):

Topology	KEEP	K50	K100	K200	K300	K500	K800	Most used
Abilene	607	983	0	0	4	51	371	K50
GEANT	0	5	0	664	0	1	2	K200
CERNET	92	63	3	6	36	0	0	KEEP
Sprintlink	36	0	0	0	4	20	140	K800
Tiscali	136	0	0	0	1	16	47	KEEP
Ebone	185	15	0	0	0	0	0	KEEP
Germany50	0	0	0	0	0	0	288	K800
VtlWavenet	1	39	0	0	0	0	0	K50

One topology uses multiple actions because the action is chosen per traffic-matrix cycle, not once per topology; different TMs need different K. KEEP can dominate while mean K is nonzero because KEEP reuses the carried routing between optimize cycles.

6. RL policy and reward settings

Parameter	Value
Controller	Double-DQN (argmax-Q at inference; not a reward gate / not RandomForest)
Action space	KEEP, K50, K100, K200, K300, K500, K800
gamma (discount)	0.5
W_PR (PR reward)	10.0
W_MLU	5.0
W_DB (lambda for DB)	20.0
W_MS (decision-time)	0.003
W_K (K/active penalty)	0.5
target bonus / gate	+10 if PR>=target; strong penalty if PR<target; flat anti-KEEP-below-target; ms>500 gate
epsilon	1.0 -> 0.05 over training
episodes	22 (x6 seen topos x 160 cycles)

7. Zero-shot generalization

Zero-shot topology	N	Mean PR	PR>=.90	PR>=.95	Min PR	Mean DB	P95 DB	Mean ms
Germany50	288	0.9878	97%	96%	0.4090	0.0098	0.0406	212.6
VtlWavenet	40	0.9373	100%	0%	0.9310	0.0007	0.0014	295.9
Combined zero-shot	328	0.9816	98%	84%	0.4090	0.0087	0.0319	222.8

Germany50 and VtlWavenet were never seen in training; both achieve PR>=0.90 with mean decision time <500 ms.

8. Decision-time breakdown, solver, and LP size

Decision-time components (GNN/DDQN scoring is a per-topology constant; LP is the remainder of optimize cycles):

Topology	GNN/scoring ms	LP ms (optimize avg)	Total mean ms	P95 ms	Max ms
Abilene	3	12.0	10.7	20.4	180.0
GEANT	7	59.6	66.6	121.1	199.2
CERNET	22	62.9	46.1	120.7	215.2
Sprintlink	27	186.0	174.8	278.8	376.0
Tiscali	33	206.0	76.8	298.4	387.4
Ebone	12	21.8	3.0	33.6	35.2
Germany50	26	186.6	212.6	285.5	375.6
VtlWavenet	140	163.4	295.9	307.3	477.2

Decision time by action type (pooled over all topologies):

Action	Rows	Mean ms	P95 ms	Max ms
KEEP	1057	0.5	0.5	0.5
K50	1105	25.7	57.7	477.2
K100	3	70.3	71.0	71.0
K200	670	67.1	121.2	199.2
K300	45	126.0	220.6	259.1
K500	88	79.4	195.9	239.7
K800	848	132.9	278.2	387.4

Solver / reproducibility:

Item	Value
LP solver	PuLP + CBC (open-source)
LP type	selected-flow path LP, DB-budgeted
k_paths	8 for K50-K300; 4 for K500/K800
Seed	42
Framework	PyTorch (CPU)

LP problem size (per topology):

Topology	Nodes	Links	OD pairs	selected OD avg	k_paths	LP vars ~avg	Mean ms
Abilene	12	30	127	51	8	~412	10.7
GEANT	22	72	425	200	8/4	~1200	66.6
CERNET	41	116	1640	77	8	~618	46.1
Sprintlink	44	166	1892	616	8/4	~3696	174.8
Tiscali	49	172	2352	230	8/4	~1377	76.8
Ebone	23	76	506	4	8	~30	3.0
Germany50	50	176	1441	799	8/4	~4795	212.6
VtlWavenet	92	192	8372	49	8	~390	295.9

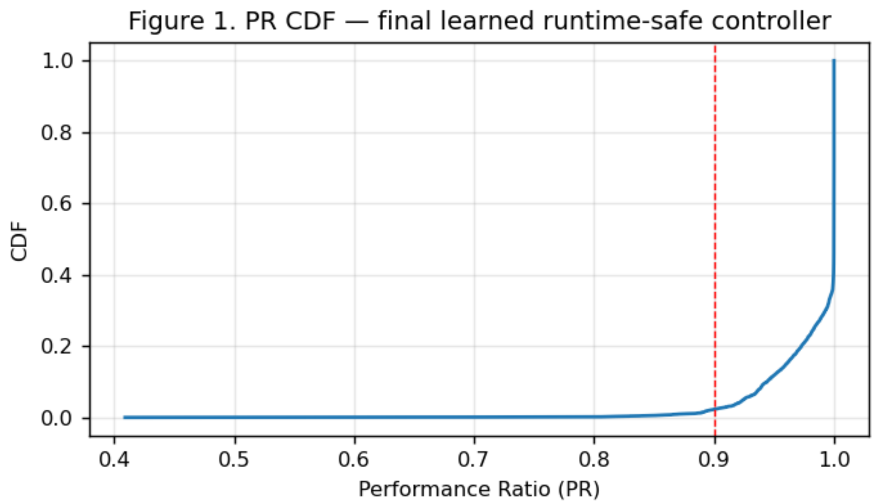
9. Failure scenarios

Failure-link scenarios (single/two/three-link failure, random link failure 1/2, capacity degradation 50%, mixed spike+failure) were NOT rerun for the frozen final controller. No failure metrics, disconnected-OD tables, or failure CDFs are claimed for the final method. The earlier offline failure artifacts in the repository belong to an earlier method and are not attributed to this controller.

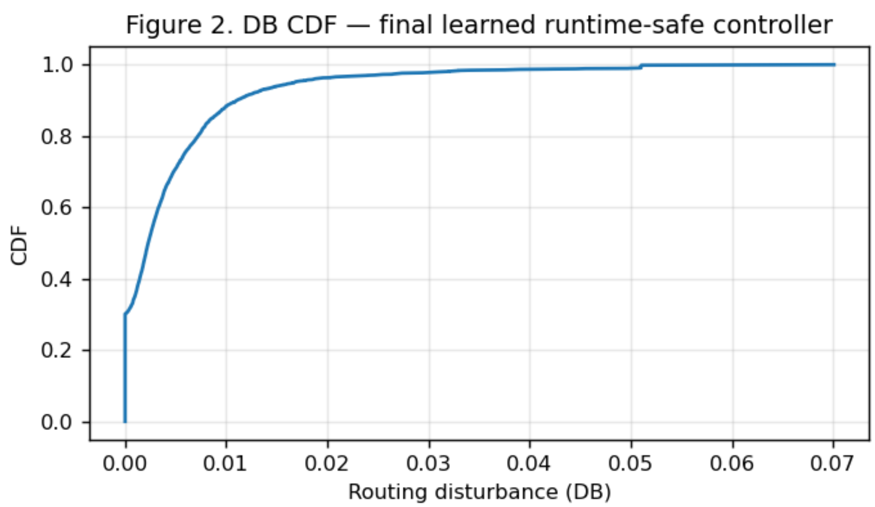
10. Mininet / SDN validation

Live SDN/Mininet metrics are retained from the prior Phase 1.5 operational-validation artifact (Abilene and GEANT, normal + failure scenarios: throughput, RTT, jitter, loss, rule count, install ms, recovery ms, disconnected). They are separate evidence and the final Iter2 DDQN was NOT rerun inside Mininet. See the main report (Section 11) for the retained table. SDN rule-installation time is reported separately from offline decision time and must not be mixed with it.

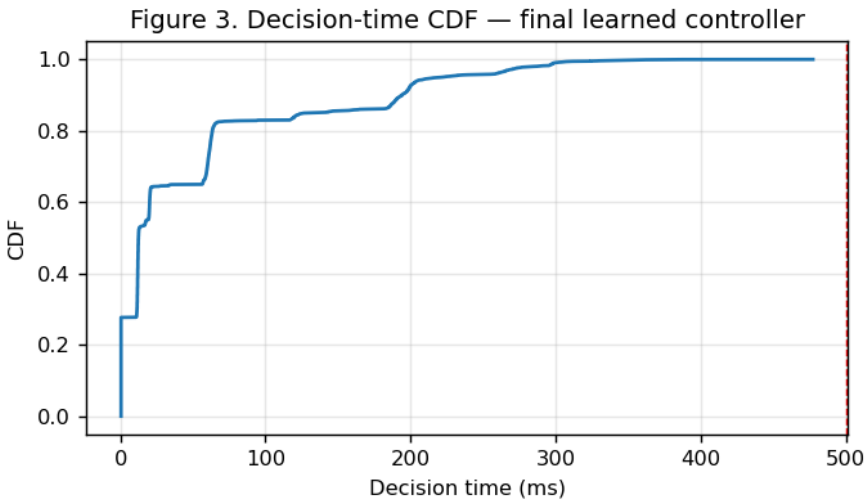
11. Figures (regenerated from final CSVs)



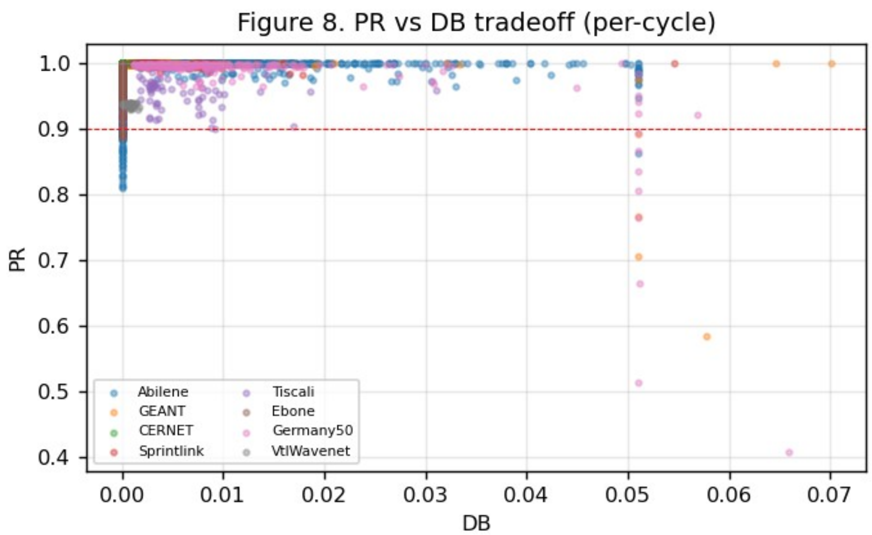
PR CDF



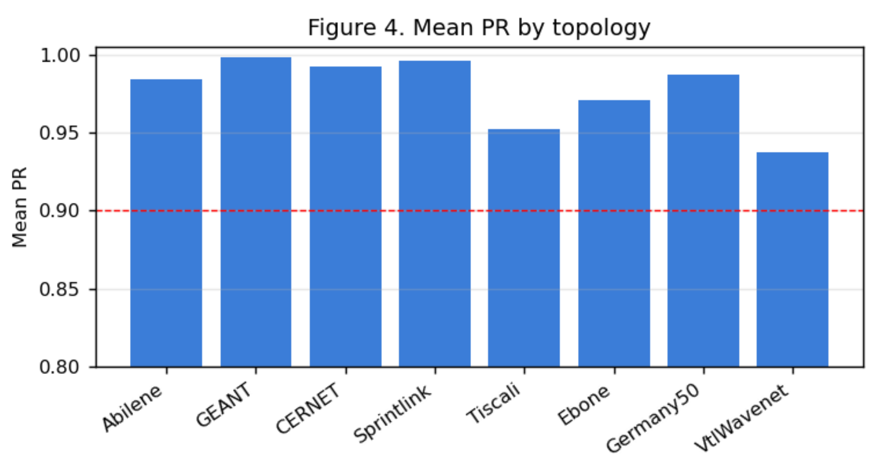
DB CDF



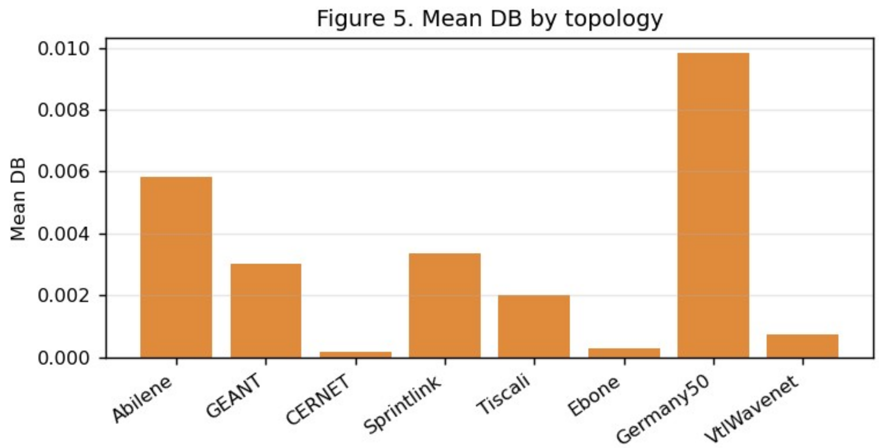
Decision-time CDF



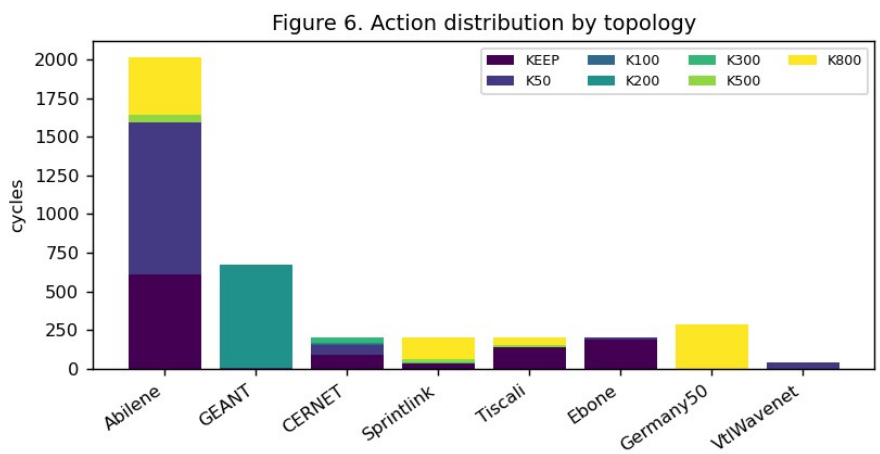
PR-DB tradeoff



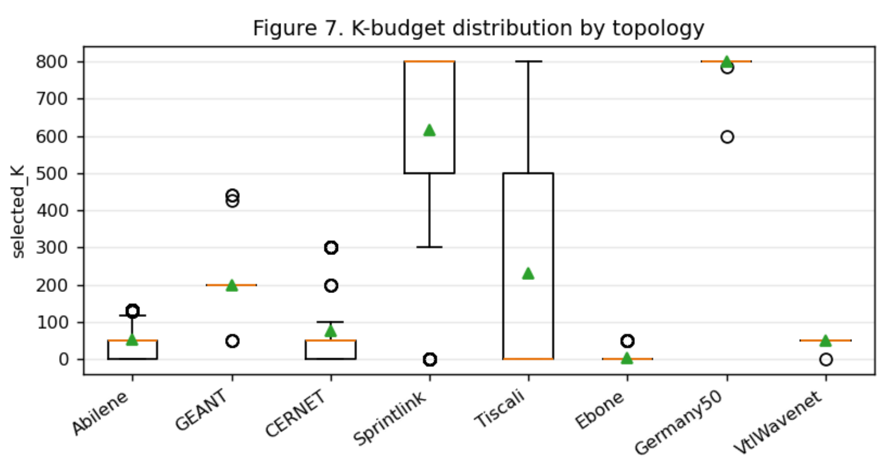
Mean PR by topology



Mean DB by topology



Action distribution



K-budget distribution

Failure normalized-MLU CDF, failure DB-by-scenario, and the Mininet throughput/loss chart are NOT regenerated for the final method because failure scenarios were not rerun and the final controller was not rerun in Mininet.

12. Reproducibility

Item	Value
Final model	final_learned_4of5_iter2_model.pt (Double-DQN, dim=33, 7 actions)
Per-cycle data	final_learned_4of5_iter2_eval_per_cycle.csv (3816 rows)
Train command	python3 scripts/phase1_5/run_final_iter2.py
Verify command	python3 verify_results.py (in the reproduction package)
Counters	td_updates=20620, target_updates=42, ce_updates=0
Seed / solver	42 / PuLP+CBC